

Time Series TA Session #2

April 2, 2021

Typo on slide #29/35

$$\gamma(h) = \begin{cases} \sigma_\varepsilon^2 \sum_{i=0}^{q-|h|} \theta_i \theta_{i+|h|} & \text{if } |h| \leq q \\ 0 & \text{if } |h| > q \end{cases}$$

$\boxed{\theta_0 = 1}$

$\theta_0 \equiv 1$?

slide 3

$$X_t = \phi_0 + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \varepsilon_t$$

$$E(X_t) = \mu = \phi_0 + \phi_1 \mu + \dots + \phi_p \mu + 0$$

$$\mu(1 - (\phi_1 + \dots + \phi_p)) = \phi_0, \quad \mu = \frac{\phi_0}{1 - \sum \phi_i}$$

AR(z) - region of weak stationarity

$$M = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} \quad \det(M - \lambda I) = \begin{vmatrix} \phi_1 - \lambda & \phi_2 \\ 1 & -\lambda \end{vmatrix}$$

$$= (\phi_1 - \lambda)(-\lambda) - \phi_2 = 0$$

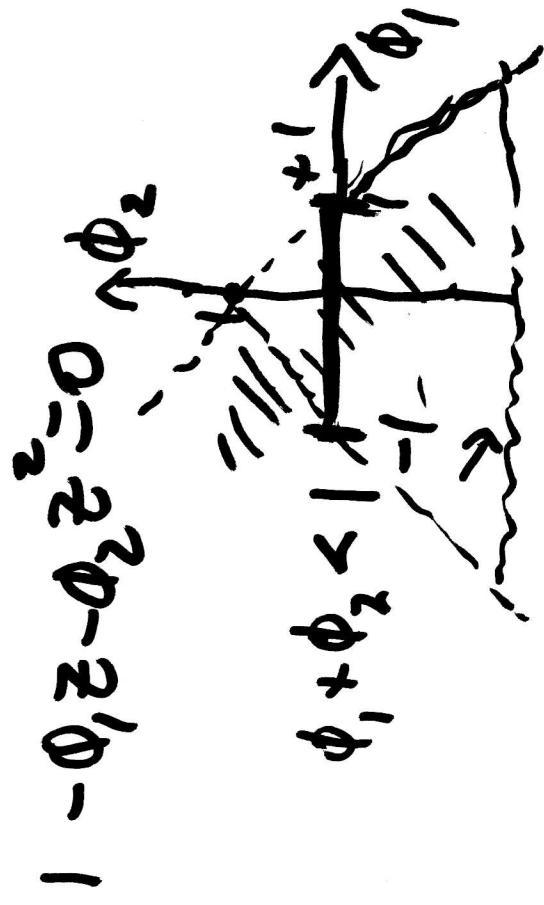
$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0$$

autoregressive polynomial $\lambda^2 - \phi_1 \lambda - \phi_2 = 0$

$$\left| \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1$$

$$|\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}| < 2$$

②



$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

$$Y_t - \mu = Y_t - \frac{\phi_0}{1 - (\phi_1 + \phi_2)}$$

$$Y_t - \mu = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

Simulation of time series models using

Statsmodels.tsa...

There is a parameter to call a function

To simulate a desired noise dist,

Question 26 check whether $\sigma_\varepsilon^2 = .5$ or $\sigma_\varepsilon = .5$?

"Random walk"

$\{X_i\}_{i=1}^{\infty}$ iid random variables

$$S_0 = 0, S_n = \sum_{i=1}^n X_i$$

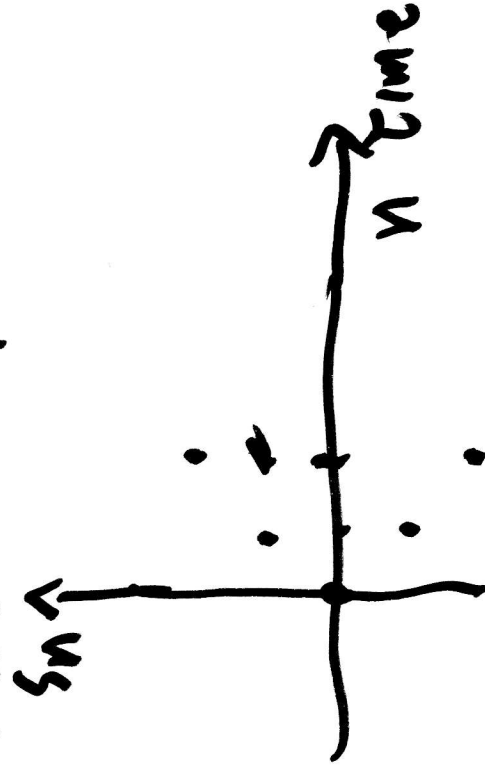
$\{S_n\}_{n=0}^{\infty}$ is a random walk

Commonly $\{X_i\}_{i=1}^{\infty}$ are iid

Also - simple random walk $\{X_i\}_{i=1}^{\infty}$ are iid ~~Bernoulli~~ with two values

$$\begin{cases} +1 & \text{wp } p \\ -1 & \text{wp } 1-p \end{cases}$$

"Random walk down Wall Street"
B. Mandelbrot



$$\text{AR}(2) \quad X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t \quad \mu = 0$$

Find conditions on (ϕ_1, ϕ_2) for weak stationarity

$$M = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} \quad \text{Eigenvalues: } \det(\phi_1 - \lambda \quad \phi_2 \\ 1 \quad -\lambda) = 0$$

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0$$

$$\lambda = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}, \quad \text{need } |\lambda| < 1$$

2 Cases: λ real or λ complex ($\phi_1^2 + 4\phi_2 < 0$ or $\phi_1^2 < -4\phi_2$)

if λ is real $-2 < \phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} < 2$

$$\Leftrightarrow \sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1, \quad \text{or } \phi_1^2 + 4\phi_2 < 4 - 4\phi_1 + \phi_1^2$$

$$\Leftrightarrow \phi_1 + \phi_2 < 1$$

Similarly $-2 - \phi_1 < -\sqrt{\phi_1^2 + 4\phi_2}$

or $\sqrt{\phi_1^2 + 4\phi_2} < 2 + \phi_1 \Rightarrow \phi_1^2 + 4\phi_2 < 4 + 4\phi_1 + \phi_1^2$

$\Rightarrow \phi_2 < 1 + \phi_1$

In the complex λ case ($\phi_1^2 < -4\phi_2$)

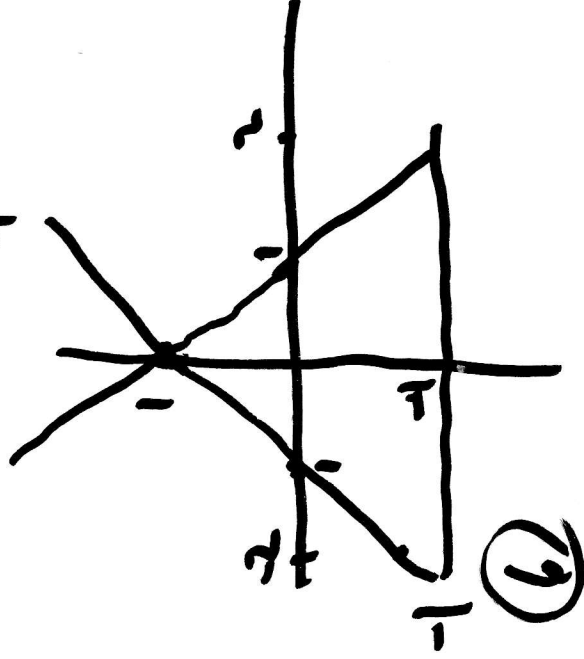
$$\lambda = \frac{\phi_1}{2} \pm i \frac{\sqrt{-(\phi_1^2 + 4\phi_2)}}{2}$$

, the modulus must be < 1 , so the squared modulus must be less than 1

$$\therefore \frac{\phi_1^2}{4} - \frac{(\phi_1^2 + 4\phi_2)}{4} = -\phi_2$$

need $-\phi_2 < 1$

or $\phi_2 > -1$



$\therefore (\phi_1, \phi_2)$ must be below the ~~lines~~ lines

$\phi_1 + \phi_2 < 1$

$\phi_2 < 1 + \phi_1$

and $\phi_2 > -1$

The triangular region

(1)